Z-transformacija nekog sustava – općenito

$$y[n] + a_1 y[n-1] + \dots + a_N y(n-N) = b_0 \cdot u[n] + b_1 u[n-1] + \dots + b_N u[n-N]$$

$$Y(z) + a_1 \cdot \left[z^{-1} \cdot y[z] + y[-1]\right] + \dots + a_N \cdot \left[z^{-N} \cdot y[z] + z^{-N} \sum_{m=-N}^{-1} y[m] z^{-m}\right]$$

$$= b_0 \cdot u(z) + b_1 \cdot \left(z^{-1} \cdot u(z) + b \cdot u[-1]\right) + \dots + b_N \left[z^{-N} \cdot U(z) + z^{-N} \sum_{m=-N}^{-1} u(m) \cdot z^{-m}\right]$$

$$Y(z)\left(1 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_N \cdot z^{-N}\right) + \left[a_1 y(-1) + \dots + a_N z^{-N} \sum_{m=-N}^{-1} y(m) z^{-m}\right]$$

$$= U(z)\left(b_0 + b_1 \cdot z^{-1} + \dots + b_N z^{-N}\right) + \left[b_1 u(-1) + \dots + b_N z^{-N} \sum_{m=-N}^{-1} u(m) \cdot z^{-m}\right]$$

$$Y(z) = \frac{b_0 + b_1 z^{-1} + \dots + b_N z^{-N}}{1 + a_1 z^{-1} + \dots + a_N \cdot z^{-N}} U(z) + \underbrace{\frac{b_1 \cdot u(-1) + \dots - (a_1 y(-1) + \dots)}{1 + a_1 z^{-1} + a_2 z^{-2} + \dots + a_n z^{-N}}}_{\text{ODZIV NEPOBUĐENOG SUSTAVA}}$$

ODZIV MIRNOG SUSTAVA

$\Rightarrow$ totalni odziv

* U slučaju impulsnog odziva: na ulazu impuls: $u[n] = \delta[n]$

$$u[n] = \delta[n] \longrightarrow U(z) \longrightarrow Y(z) = \frac{b_0 + b_1 \cdot z^{-1} + \dots + b_N \cdot z^{-N}}{1 + a_1 \cdot z^{-1} + \dots + a_N \cdot z^{-N}}$$

$$y[n] \Rightarrow h[n] \longrightarrow H(z)$$

prijenosna funkcija u Z domeni

Primjer:

$$y[n] - 0{,}9 y[n-1] + 0.2 y[n-2] = u[n] + u[n-1] + u[n-2]$$

! kada radimo sa Z-transf. i mi smo pretpostavili da su naši signali kauzalni. Z-transf možemo napraviti samo ako je ulaz pomnožen sa stepom!

$y[-1] = 6$

$y[-2] = -3$

$u[n] = -9(-0{,}9)^n \mu[n]$

u Z-transf ne moramo preračunavati poč uvjete!

$$Y(z) - 0.9\left(z^{-1} Y(z) + Y(-1)\right) + 0.2\left(z^{-2} \cdot Y(z) + Y(-2) \cdot z^{-1} + Y(-1)\right) =$$

$$= U(z) + \left(z^{-1} \cdot U(z) + \cancel{u(-1)}_{0}\right) + \left[z^{-2} U(z) + \cancel{u(-2)}_{0} + \cancel{u(-1)}_{0}\right]$$

uvrstimo poč. uvjete:

$$Y(z) - 0{,}9 \cdot z^{-1} \cdot Y(z) - \underline{0{,}9 \cdot y(-1)} + 0{,}2\left[z^{-2} \cdot Y(z)\right] + \underline{0{,}2\left(y(-2) + y(-1) \cdot z^{-1}\right)}$$

$$= U(z)\left(1 + z^{-1} + z^{-2}\right)$$

$$Y(z)\left(1 - 0{,}9z^{-1} + 0{,}2z^{-2}\right) \quad (\cdot +0) \quad = U(z)\left(1 + z^{-1} + z^{-2}\right)$$

$$Y(z) = \underbrace{\frac{1 + z^{-1} + z^{-2}}{1 - 0{,}9z^{-1} + 0{,}2z^{-2}} U(z)}_{\text{samo ulaz ulaze}} + \underbrace{\frac{0{,}9\, y(-1) - 0{,}2\left(y(-2) + y(-1)\, z^{-1}\right)}{1 - 0{,}9z^{-1} + 0{,}2z^{-2}}}_{\text{kad ga vratimo u izlaz (ovo ne ovisi o ulazu, samo o poč. uvjetu izlaza)}}$$

$\Rightarrow$ MIRNI SUSTAV

TOTALNI ODZIV

$\Rightarrow$ NEPOBUĐENI SUSTAV

$$y(z) = H(z) \cdot U(z) \longrightarrow H(z) = \frac{Y(z)}{U(z)} \longrightarrow h[n]$$

$$u[n] = -9 \cdot (-0{,}9)^n \mu[n] \longrightarrow U(z) = -9 \cdot \frac{z}{z + 0{,}9}$$

Znači odziv mirnog sustava je onda: $Y_m(z) = H(z) \cdot U(z)$

$$Y_m(z) = \frac{1 + z^{-1} + z^{-2}}{1 - 0{,}9z^{-1} + 0{,}2z^{-2}} \cdot (-9) \cdot \frac{z}{z + 0{,}9} = \frac{z^2 + z + 1}{z^2 - 0{,}9z + 0{,}2} \cdot \frac{-9z}{z + 0{,}9}$$

da bismo vratili u vremenski domenu, moramo rastaviti na parc. razlomke i tražiti u tablici

$$Y_m(z) = \frac{(z^2 + z + 1)(-9 \cdot z)}{(z - 0{,}5)(z - 0{,}4) \cdot (z + 0{,}9)}$$

1. sljedeći korak ne postoji kod Laplacea! Ne mijenjaj!

podijelimo sa z: $Y_{m_1}(z) = \dfrac{z^2 + z + 1}{(z - 0{,}5)(z - 0{,}4)(z + 0{,}9)} \cdot (-9)$ i to sad faktoriziramo

$$Y_{m_1} = \frac{A}{z - 0{,}5} + \frac{B}{z - 0{,}4} + \frac{C}{z + 0{,}9}$$

$\longrightarrow$ kada nađemo sve, onda POMNOŽIMO sve sa Z

$$y_{m_1} = \frac{A}{z-0{,}5} + \frac{B}{z-0{,}4} + \frac{C}{z+0{,}9}$$

1° $C_i = \lim_{z \to p_i} \{(z-p_i) \cdot Y(z)\}$

$$A = \lim_{z \to 0{,}5} \cancel{(z-0{,}5)} \cdot \frac{-9z^2-9z-9}{\cancel{(z-0{,}5)}(z-0{,}4)(z+0{,}9)} = \lim_{z \to 0{,}5} \frac{-9z^2-9z-9}{(z-0{,}4)(z+0{,}9)}$$

$$A = -112{,}5$$

$$B = \lim_{z \to 0{,}4} \cancel{(z-0{,}4)} \frac{-9z^2-9z-9}{(z-0{,}5)\cancel{(z-0{,}4)}(z+0{,}9)} = \dots = 108$$

$$C = \lim_{z \to -0{,}9} \cancel{(z+0{,}9)} \frac{-9z^2-9z-9}{(z-0{,}5)(z-0{,}4)\cancel{(z+0{,}9)}} = -4{,}5$$

$$y_{m_1}(z) = \frac{-112{,}5 \cdot z}{z-0{,}5} + \frac{108 \cdot z}{z-0{,}4} - \frac{4{,}5 \cdot z}{z+0{,}9}$$

$$y_m(z) = -112{,}5 \frac{z}{z-0{,}5} + 108 \frac{z}{z-0{,}4} - 4{,}5 \cdot \frac{z}{z+0{,}9}$$

$$\boxed{y_m[n] = \left[-112{,}5 \cdot 0{,}5^n + 108 \cdot 0{,}4^n - 4{,}5 \cdot (-0{,}9)^n\right] \mu[n]}$$

ne zaboravi!

ODZIV MIRNOG SUSTAVA

2° ODZIV NEPOBUĐENOG

$$y_0(z) = \frac{0.9 \cdot y(-1) - 0{,}2(y(-2) + y(-1)z^{-1})}{1 - 0{,}9z^{-1} + 0.2z^{-2}} \Bigg/ \cdot z^2 = \frac{6z^2 - 1.2z}{z^2 - 0{,}9z + 0{,}2}$$

$$y_0(z) = \frac{z(6z - 1.2)}{(z-0{,}5)(z-0{,}4)}$$

$$y_{01}(z) = \frac{6z - 1.2}{(z-0{,}5)(z-0{,}4)} = \frac{A}{(z-0.5)} + \frac{B}{(z-0.4)}$$

* Novi način je bio i prije primjenjivan pa tako je i ovaj na gore

$$\Rightarrow \frac{A(z-0.4) + B(z-0.5)}{(z-0{,}5) \cdot (z-0{,}4)} = \frac{6z - 1{,}2}{(z-0.5)(z-0{,}4)}$$

$$A + B = 6$$

$$A(-0{,}4) + B \cdot (-0{,}5) = -1.2$$

$$B = -12$$

$$A = 18$$

$$y_{01}(z) = \frac{18}{z-0{,}5} - \frac{12}{z-0{,}4}$$

$$y_0 = 18 \frac{z}{z-0{,}5} - 12 \cdot \frac{z}{z-0{,}4} \Rightarrow \boxed{y_0[n] = \left[18 \cdot 0{,}5^n - 12 \cdot 0{,}4^n\right] \mu[n]}$$

ODZIV NEPOBUĐENOG SUSTAVA

$$\boxed{y[n] = y_m[n] + y_0[n] = \left[-94.5 \cdot 0{,}5^n + 96 \cdot 0{,}4^n - 4.5 \cdot (-0{,}9)^n\right] \mu[n]}$$

1. pomoću kašnjenja odredi Z domenu

2.1 Odnosno možemo gledati MIRNI i NEPOBUĐENI

2.2 Ako nas pita totalni možemo u U(z) direktno uvrstiti

3 Ako kaže da je sustav MIRAN → ne moramo uvrštavati poč uvjete, samo $H(z)\cdot U(z)$

Primjer: $y[n]+\frac{1}{2}y[n-1]-\frac{1}{4}y[n-2]-\frac{1}{8}y[n-3]=8u[n-3]$

tražimo samo impulsni odziv

$u[n]=\delta[n] \longrightarrow U(z)=1$ zbog dirac → sve prije $\emptyset \to \emptyset$

$$Y(z)+\frac{1}{2}\left(z^{-1}\cdot Y(z)+\cancel{y(-1)}\right)-\frac{1}{4}\left(z^{-2}Y(z)+\cancel{y(-2)}+\cancel{y(-1)}\cdot z^{-1}\right)$$

$$-\frac{1}{8}\left(z^{-3}Y(z)+\cancel{y(-3)}+\cancel{y(-2)}+\cancel{y(-1)}\cdot z^{-1}\right)=8\cdot z^{-3}\cdot U(z)$$

$$Y(z)+\frac{1}{2}z^{-1}\cdot Y(z)-\frac{1}{4}z^{-2}\cdot Y(z)-\frac{1}{8}z^{-3}Y(z)=8\cdot z^{-3}U(z)$$

$$Y(z)\left(1+\frac{1}{2}z^{-1}-\frac{1}{4}z^{-2}-\frac{1}{8}z^{-3}\right)=8z^{-3}\cdot U(z)$$

$$Y(z)=\frac{8z^{-3}}{1+\frac{1}{2}z^{-1}-\frac{1}{4}z^{-2}-\frac{1}{8}z^{-3}}\cdot U(z)$$

$$(8z^{-3}):\left(-\frac{1}{8}z^{-3}-\frac{1}{4}z^{-2}+\frac{1}{2}z^{-1}+1\right)=-64$$

$$-8z^{3}-16z^{2}+32z^{-1}+64$$

$$-16z^{2}+32z^{-1}+64$$

$$H(z)=-64+\frac{-6z^{-2}+32z^{-1}+64}{1+\frac{1}{2}z^{-1}-\frac{1}{4}z^{-2}-\frac{1}{8}z^{-3}}$$

$$H_1(z)=\frac{-16z+32z^{2}+64z^{3}}{z^{3}+\frac{1}{2}z^{2}-\frac{1}{4}z-\frac{1}{8}}=\frac{64z^{3}+32z^{2}-16z}{(z-0{,}5)(z+0{,}5)^{2}}$$

svaki član ima z, taman za podijeliti!

$$H_2(z) = \frac{H_1(z)}{z} = \frac{64z^2+32z-16}{(z-0{,}5)(z+0{,}5)^2} = \frac{A}{z-0{,}5} + \frac{B}{z+0{,}5} + \frac{C}{(z+0{,}5)^2}$$

biramo način ---> dobije se $A=16$

$$C_i = \frac{1}{(r-j)!} \cdot \lim_{z\to p_i} \left\{ \frac{d^{r-j}}{dz^{r-j}} (z-p_i)^r \cdot X(z) \right\} \Rightarrow$$ za višestruke polove

najveća potencija našeg nazivnika „j" je trenutna potencija
↳ ovdje => $r=2$

$$B = \frac{1}{\cancel{(2-1)!}} \lim_{z\to -\frac{1}{2}} \left\{ \frac{d^{2-1}}{dz^{2-1}} \cancel{\left(z+\frac{1}{2}\right)^2} \cdot \frac{64z^2+32z-16}{(z-0{,}5)\cancel{(z+0{,}5)^2}} \right\}$$

$$B = \lim_{z\to -\frac{1}{2}} \frac{(64\cdot 2\cdot z+32)(z-0{,}5)-(64z^2+32z-16)\cdot 1}{\left(z-\frac{1}{2}\right)^2} = \dots \Rightarrow B=48$$

$$C = \frac{1}{\cancel{(2-2)!}} \lim_{z\to -\frac{1}{2}} \left\{ \cancel{\frac{d^{2-2}}{dz^{2-2}}} \cancel{\left(z+\frac{1}{2}\right)^2} \cdot \frac{64z^2+32z-16}{(z-0{,}5)\cancel{(z+0{,}5)^2}} \right\}$$

$$C = \lim_{z\to -\frac{1}{2}} \frac{64z^2+32z-16}{z-\frac{1}{2}} = \dots \Rightarrow C=16$$

$$H(z) = -64 + \left|{}^{*}(\text{množimo sa } z)\right| + \frac{16z}{z-0{,}5} + \frac{48z}{z+0{,}5} + \frac{16z}{(z+0{,}5)^2}$$

$-32\cdot\left(-\frac{1}{2}\right)$

$$h[n] = \left(-64\,\delta[n] + 16\cdot(0{,}5)^n + 48\cdot(-0{,}5)^n + \left(-32\cdot\frac{-\frac{1}{2}z}{\left(z+\frac{1}{2}\right)^2}\right)\right)\mu[n]$$

$^{*}\alpha = -\frac{1}{2}$

$$\boxed{h[n] = -64\,\delta[n] + \left(16\cdot(0{,}5)^n + 48\cdot(-0{,}5)^n - 32\cdot n\,\frac{-\frac{1}{2}z}{\left(z+\frac{1}{2}\right)^2}\right)\mu[n]}$$